

Scope & Reality

Back in week one, my original idea was to make a split, ergonomic keyboard that had mechanical switches, but the final product was completely different as it was a macropad but still had mechanical switches. The original idea for a full keyboard was too ambitious and was going to be too difficult for me to successfully design and build. So, instead of doing that, I decided to make a macropad because it was simpler to make but was still a keyboard fundamentally. For the first iteration of my macropad, I wanted to design and manufacture a custom PCB so that the assembly would be much simpler to put together and design. However, while trying to learn PCB design in KiCAD it became too tedious, and I decided to just hardwire all of the switches, the screen, RGB strip, and rotary encoders. The overall design changed completely from my original week one idea, but fundamentally, they stayed the same.

Skill Acquisition

While making this macropad, I gained a large multitude of new skills and all of them are going to aid me in all my future projects. The hardest skill that I had to learn to make this project work was learning how to program in Circuit Python and how to use a Raspberry PI Pico because when it comes to doing electronics with a microcontroller, I am used to using an Arduino brand board and then programming them in C++. It was difficult to pivot to the new programming language and then learn how to upload the code to the Pico. Another difficult skill I had to master was precision soldering because in the v1 macropad, I made the columns and rows of the keypad out of enameled copper wire that was thinner than a human hair (which wasn't a great idea in hindsight) and had to use a blunt tipped soldering iron to solder the wire to the pins of the mechanical switches. It was difficult to do but was easy to learn and this skill is going to be extremely crucial when assembling PCBs.

Next Steps

The end product was a good start, but there are a lot of ways that I could have improved if I had more time and money. If I had 2 more months and \$120 extra dollars, I could have designed a custom PCB and could have switched the microcontroller from a Raspberry PI Pico to an Arduino (Pro) Micro or a different HID compatible microcontroller that

could be programmed with C++. The PCB would have helped tremendously with assembling the electronics because there wouldn't have had to be so many wires, and hot swappable switches like I originally planned would have been possible. The only skill that I would have needed to learn was surface mounted soldering if I had made the PCB. With this current product, it is a good starting point for any future improvement that I might make in my own time. Overall, the V3 macropad would have much cleaner electronics, a user-friendly UI, and maybe even Bluetooth connectivity.